



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

UNIT TEST

30-05-2021

JEE(Main + Advance) : Nurture (PHASE : I,I(B),II(A) & NURTURE (LIVE) I,I(B),II(A))

Test Type : UNIT TEST

Test Pattern : JEE (Main)

TEST DATE : 30-05-2021

ANSWER KEY

PHYSICS

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-A	A.	A	A	C	C	C	B	B	B	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
SEC-B	A.	C	B	C	C	A	D	B	C	B	D
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	259.00	7.00	1.00	0.60	0.00	63.00	5.00	3.00	22.00	4.00

CHEMISTRY

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-A	A.	B	B	C	D	A	A	C	A	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
SEC-B	A.	B	B	C	A	C	B	D	D	D	D
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4.00	6.00	4.00	2.00	0.40	1.25	6.00	4.00	2.00	3.00

MATHEMATICS

	Q.	1	2	3	4	5	6	7	8	9	10
SEC-A	A.	C	C	B	B	A	C	A	A	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
SEC-B	A.	A	C	A	B	B	D	B	D	A	A
	Q.	1	2	3	4	5	6	7	8	9	10
	A.	-12.00	1.25	2.00	-1.00	0.00	0.00	8.00	3.00	6.00	0.00

HINT – SHEET

PHYSICS

SEC-A

1. Ans (A)

For vectors to be perpendicular,

$$\vec{A} \cdot \vec{B} = 0 \Rightarrow a = -4$$

2. Ans (A)

$$\frac{P+q}{P-q} = \frac{3}{1}$$

$$\Rightarrow P = 2q$$

3. Ans (C)

$$|\hat{A}| = 1 \Rightarrow \sqrt{0.5^2 + 0.8^2 + C^2} = 1$$

$$\Rightarrow C = \sqrt{0.11}$$

4. Ans (C)

$$\vec{r}_i = \hat{i}$$

$$\vec{d}_1 = 21 \times 5 \left[\frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{7} \right]$$

$$= 30\hat{i} + 45\hat{j} - 90\hat{k}$$

$$\vec{d}_2 = 5\sqrt{2} \times 5 \frac{[3\hat{i} + 4\hat{j} + 5\hat{k}]}{5\sqrt{2}}$$

$$= 15\hat{i} + 20\hat{j} + 25\hat{k}$$

$$\vec{d}_{Net} = 45\hat{i} + 65\hat{j} - 65\hat{k}$$

$$\vec{r}_f = 46\hat{i} + 65\hat{j} - 65\hat{k}$$

$$\vec{d}_f = -\vec{r}_f = -(46\hat{i} + 65\hat{j} - 65\hat{k})$$

$$= -46\hat{i} - 65\hat{j} + 65\hat{k}$$

5. **Ans (C)**

Here $\vec{a} \cdot \vec{b} < 0$ so angle between them is obtuse.

$$\vec{a} \cdot \vec{c} = (-3\hat{i} + 2\hat{j} - \hat{k}) \cdot (2\hat{i} + \hat{j} - 4\hat{k})$$

$$= -6 + 2 + 4 = 0$$

$$|\vec{a}| = \sqrt{14}, |\vec{b}| = \sqrt{35}, |\vec{c}| = \sqrt{21}$$

$$\Rightarrow b^2 = a^2 + c^2$$

6. **Ans (B)**

$$0 = (100)^2 - 2 \times a \times 0.1$$

$$a = \frac{10^4 \times 10}{2} = 5 \times 10^4$$

$$F = 10 \times 10^{-3} \times 5 \times 10^4$$

$$F = 500 \text{ N}$$

7. **Ans (B)**

Force is in positive direction where acceleration is positive i.e. slope of v-t graph is positive.

8. **Ans (B)**

Here acceleration = constant. Therefore body reaches the farthest point after $20/2=10$ s and then starts coming back towards the initial point. Hence position of body at $13s=(10+3)$ s will be same as at $(10-3) = 7$ s

9. **Ans (A)**

Let both particles reach at same position in same time t then from $s = ut + \frac{1}{2} at^2$

$$\text{For 1st particle : } s = 4(t) + \frac{1}{2} (A) t^2 = 4t + \frac{t^2}{2}$$

$$\text{For 2nd particle : } s = 2(t) + \frac{1}{2} (B) t^2 = 2t + \frac{t^2}{2}$$

$$\text{Equating above equation we get } 4t + \frac{t^2}{2}$$

$$= 2t + \frac{t^2}{2} \Rightarrow t = 4 \text{ sec}$$

Substituting value of t in above equation

$$s = 4(D) + \frac{1}{2} (A) (D)^2 = 16 + 8 = 24 \text{ m}$$

10. **Ans (C)**

Ball strikes towers $\Rightarrow S_x = 0$

$$\therefore 0 = 12 \times t + \frac{1}{2} (-6) t^2 \Rightarrow t = 4 \text{ sec}$$

So displacement in y-direction in 4 sec

$$S_y = \frac{1}{2} \times 10 \times 16 = 80 \text{ m}$$

11. **Ans (C)**

Suppose particle is projected with a velocity u .

$$D = \frac{u^2}{2g}$$

$$u^2 = 2gD \quad \dots (i)$$

$$\& T = \frac{u}{g} \quad \dots (ii)$$

If there were no gravity, particle will move with constant velocity u .

$$\text{So distance} = uT = \frac{u \cdot u}{g} = \frac{u^2}{g} = \frac{u^2}{g} = 2D$$

12. **Ans (B)**

for first one second $v = 10 \times 1 = 10 \text{ m/s}$

$$h = \frac{1}{2} gt^2 = 5 \text{ m}$$

distance travelled in next 5sec = $10 \times 5 = 50 \text{ m}$

total time = 8 sec

time left = 2 sec

$$\text{distance in next 2 sec} = 10 \times 2 + \frac{1}{2} (0 \times 2) =$$

40 m

Total height = 95 m

14. **Ans (C)**

$$H_1 = H_2$$

$$\frac{u_1^2 \sin^2 \theta_1}{2g} = \frac{u_2^2 \sin^2 \theta_2}{2g}$$

$$u_1 \sin \theta_1 = u_2 \sin \theta_2$$

$$\text{Hence } T_1 = T_2$$

15. **Ans (A)**

$$R = 5 \left(\frac{2u \sin \theta}{g} \right)^2 = \frac{u^2 \sin 2\theta}{g}$$

$$\Rightarrow \frac{20 \sin^2 \theta}{g} = 2 \sin \theta \cos \theta$$

$$\Rightarrow \tan \theta = 1$$

16. **Ans (D)**

$$R \propto u_x \times u_y$$

17. **Ans (B)**

Horizontal component of velocity is decreasing with time.

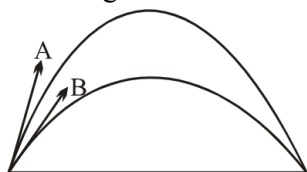
18. **Ans (C)**

$$80 = \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 4 \text{ sec}$$

$$10 \times 4 + v \times 4 = 60 \Rightarrow v = 5 \text{ m/s}$$

19. **Ans (B)**

$$T = \frac{2u \sin \theta}{g}$$



$$\theta_A > \theta_B$$

$$\therefore T_A > T_B$$

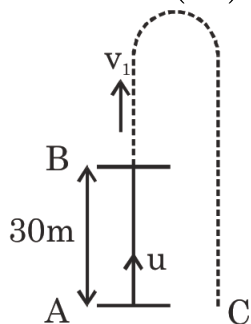
So arrow B arrives first.

20. **Ans (D)**

$$A \rightarrow B$$

$$V_1^2 - u^2 = 2as$$

$$V_1^2 - 0^2 = 2 \left(\frac{2g}{3} \right) 30$$



$$V_1 = 20 \text{ m/s}$$

$$B \rightarrow C$$

$$u = v_1 = 20$$

$$a = -g = -10$$

$$s = -30$$

$$s = ut + \frac{1}{2} at^2$$

$$t = 2 + \sqrt{10}$$

PHYSICS

SEC-B

1. **Ans (259.00)**

$$\vec{r}_f = \vec{r}_i + \vec{v}_t = (4\hat{i} + 3\hat{j} + 2\hat{k}) + \frac{(2\hat{i} + 2\hat{j} + \hat{k})}{3} \times 30 \times 5$$

$$= (104)\hat{i} + (103)\hat{j} + 52\hat{k}$$

$$x + y + z = 259$$

2. **Ans (7.00)**

$$W = \vec{F} \cdot \vec{S} = (\hat{i} + 2\hat{j})(3\hat{i} + 2\hat{j}) = 3 + 4 = 7$$

3. **Ans (1.00)**

$$\vec{A} = 3\hat{i} + 4\hat{j}$$

$$\vec{B} = 7\hat{i} + 24\hat{j}$$

$$\vec{A}_{||} = (A \cos \theta)\hat{B}$$

$$= \frac{21 + 96}{\sqrt{7^2 + 24^2}} \times \frac{7\hat{i} + 24\hat{j}}{\sqrt{7^2 + 24^2}}$$

$$= \frac{117}{625}(7\hat{i} + 24\hat{j})$$

4. **Ans (0.60)**

$$E_k = \frac{1}{2} M |\vec{v}|^2 = \frac{1}{2} M (v_x^2 + v_y^2 + v_z^2)$$

$$= \frac{1}{2} (0.1 \text{ kg}) (2^2 + 2^2 + 2^2) \text{ m}^2/\text{s}^2 = 0.6 \text{ J}$$

5. **Ans (0.00)**

$$v = 4x^2 - 4$$

$$\frac{dv}{dx} = 8x - 4$$

$$\text{at } x = 1$$

$$\frac{dv}{dx} = 4$$

$$\text{at } x = 1$$

$$v = 0$$

$$a = v \frac{dv}{dx} = 0 \times 4 = 0$$

6. **Ans (63.00)**

$$x = [(31.5) \cos 37^\circ]t = [(31.5) \cos 68^\circ]t$$

$$\text{and } y = [(31.5) \sin 37^\circ]t - \frac{1}{2}gt^2$$

$$= [(31.5) \sin 68^\circ]t - \frac{1}{2}gt^2$$

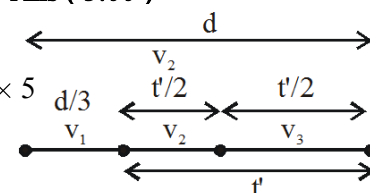
$$\text{On solving, } x = 63 \text{ m}$$

7. **Ans (5.00)**

$$\frac{dv}{dt} = 3At^2 - B = 0$$

$$t = \sqrt{\frac{B}{3A}}$$

8. **Ans (3.00)**



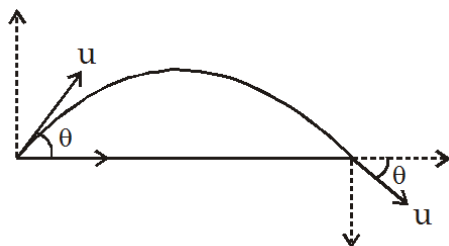
$$\frac{2d}{3} = v_2 \times \frac{t'}{2} + v_3 \times \frac{t'}{2}$$

$$t' = \frac{4d}{3(v_2 + v_3)}$$

$$v_{\text{avg}} = \frac{d}{\frac{d}{3v_1} + \frac{4d}{3(v_2 + v_3)}}$$

$$v_{\text{avg}} = \frac{3v_1(v_2 + v_3)}{4v_1 + v_2 + v_3}$$

9. Ans (22.00)



It will strike with same speed.

10. Ans (4.00)

$$-20 = 40 \tan \theta - \frac{1}{2} \frac{g \times 40^2}{u^2 \cos^2 \theta}$$

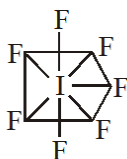
$$u^2 = \frac{1600}{4} = 400$$

$$u = 20 \text{ m/s}$$

CHEMISTRY

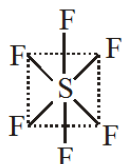
SEC-A

16. Ans (B)



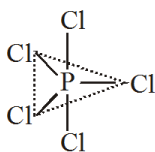
6 atom in one plane

No. of such planes 1



5 atom in one plane

No. of such planes 3



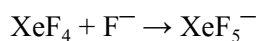
4 atom in one plane

No. of such planes 4

Option (B) is correct



17. Ans (D)



(A) Is correct as more d-orbitals are involved in hybridisation in product.

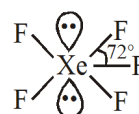
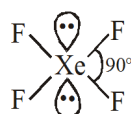
(B) Correct reactant → product



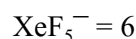
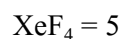
$$\approx 16\% \quad \approx 14.28\%$$

(C) Correct Square planer for XeF₄

Pentagonal planer for XeF₅⁻

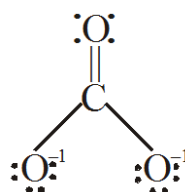


(D) Maximum atom lying in one plane.



So D option is incorrect

20. Ans (D)

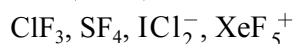


Oxidation number of C = +4

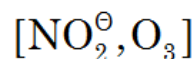
CHEMISTRY

SEC-B

3. Ans (4.00)



4. Ans (2.00)



8. Ans (4.00)

Ionisation energy of sodium, bond dissociation energy of chlorine, sublimation enthalpy of sodium, lattice energy of NaCl.

MATHEMATICS

SEC-A

1. **Ans (C)**

$$b = \frac{\alpha^2 + \beta^2}{\alpha\beta} = 8$$

second equation becomes

$$2x^2 + 10x + 24 = 0$$

$D < 0 \Rightarrow$ imaginary roots

2. **Ans (C)**

$$x^2 - bx + a = 0 \dots (1)$$

$$x^2 + ax - b = 0 \dots (2)$$

$$-(b+a)x + (a+b) = 0$$

$\Rightarrow x = 1$ is common root

Put $x = 1$ in eq (1)

$$1 - b + a = 0 \Rightarrow a - b = -1 \dots (3)$$

Now, P.O.R of eq(1) = a

$$\alpha_1 \cdot \beta_1 = a$$

$$1. \beta_1 = a$$

$$\Rightarrow \beta_1 = a$$

P.O.R of eq(2) = -b

$$\alpha_2 \cdot \beta_2 = -b$$

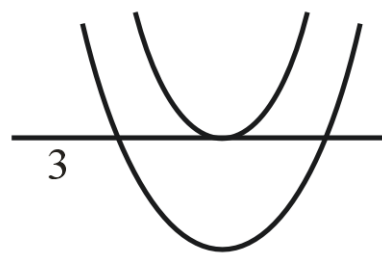
$$1. \beta_2 = -b$$

$$\Rightarrow \beta_2 = -b$$

Now, sum of uncommon roots = $a + (-b)$

$$= a - b = -1 \text{ (from (3))}$$

3. **Ans (B)**



$$f(x) = x^2 - 6ax + (2 - 2a + 9a^2)$$

$$\therefore f(3) > 0 \text{ and } D \geq 0 \text{ and } 3 < -\frac{B}{2A}$$

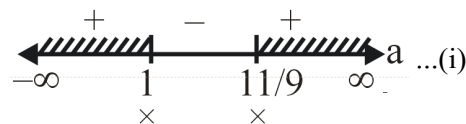
Now $f(3) > 0$

$$\Rightarrow 9 - 18a + 2 - 2a + 9a^2 > 0$$

$$\Rightarrow 9a^2 - 20a + 11 > 0$$

$$\Rightarrow 9a(a-1) - 11(a-1) > 0$$

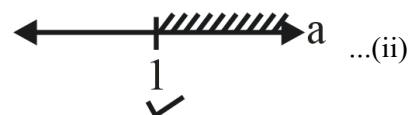
$$\Rightarrow (9a-11)(a-1) > 0$$



$$\text{Now, } D \geq 0 \Rightarrow 36a^2 - 4(1)(2 - 2a + 9a^2) \geq 0$$

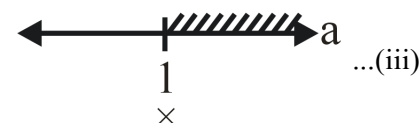
$$\Rightarrow 9a^2 - 2 + 2a - 9a^2 \geq 0$$

$$\Rightarrow a - 1 \geq 0 \Rightarrow a \geq 1$$

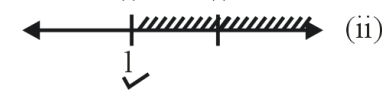
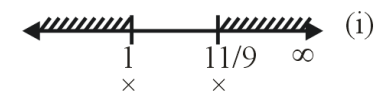


$$\text{Now } -\frac{B}{2A} > 3$$

$$\frac{6a}{2(1)} > 3 \Rightarrow a > 1$$



$$(i) \cap (ii) \cap (iii)$$



$$\therefore a \in \left(\frac{11}{9}, \infty\right)$$

4. **Ans (B)**

$$x^3 + 0x^2 + 2x - 1 = 0 \begin{matrix} \nearrow \alpha \\ \nearrow \beta \\ \nearrow \gamma \end{matrix}$$

$$* \alpha + \beta + \gamma = 0$$

$$* \alpha\beta\gamma = 1$$

$$\therefore (\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$$

$$= (-\gamma)(-\alpha)(-\beta) = -\alpha\beta\gamma = -1 \text{ Ans.}$$

5. **Ans (A)**

$$x^2 - 3x + 3 = (x - 3/2)^2 + 3/4$$

$$\therefore \text{smallest value} = 3/4 \text{ at } x = 3/2,$$

$$\text{which lie in } x \in [-3, 5/2]$$

6. **Ans (C)**

$$\left(2x + \frac{1}{x} - 3\right) \left(2x + \frac{1}{x} + 5\right) - 9 = 0$$

$$\text{Put } 2x + \frac{1}{x} = P$$

$$(P - 3)(P + 5) - 9 = 0$$

$$P = -6 \text{ or } P = 4$$

$$2x + \frac{1}{x} = -6 \text{ or } 2x + \frac{1}{x} = 4$$

$$x = \frac{-6 \pm \sqrt{28}}{4} \text{ or } x = \frac{2 \pm \sqrt{2}}{2}$$

7. **Ans (A)**

$$mx^2 - 6mx + 5m + 1 > 0 \quad x \in \mathbb{R}$$

We know that

$$\text{if } f(x) > 0 \quad \forall x \in \mathbb{R}$$

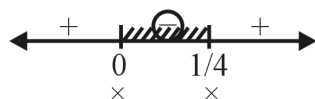
$$D < 0 \text{ and } a > 0$$

$$\therefore 36m^2 - 4(m)(5m + 1) < 0 \text{ and } m > 0$$

$$\Rightarrow 36m^2 - 20m^2 - 4m < 0$$

$$\Rightarrow 16m^2 - 4m < 0$$

$$\Rightarrow 4m(4m - 1) < 0$$



$$\therefore 0 < m < \frac{1}{4}$$

$$\text{if } m = 0$$

$$0x^2 - 6(0)x + 5 \times (0) + 1 > 0 \text{ is also true } \therefore m = 0$$

$$\therefore 0 \leq m < \frac{1}{4}$$

8. **Ans (A)**

$$a = 2, h = 2, b = -k, g = 2, f = 1, c = -1$$

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$2k + 8 - 2 + 4k + 4 = 0$$

$$6k = -10 \quad \therefore k = -\frac{5}{3}$$

9. **Ans (A)**

$$f(x) = (x - 2)(x - 4) + \lambda(x - 1)(x - 3)$$

$$f(1) = (-1)(-3) = 3 > 0$$

$$f(3) = (1)(-1) = -1 < 0$$

$$f(1)f(3) < 0$$

Roots of the given quadratic will be real and distinct

10. **Ans (C)**

$$ax^2 + bx + c = 0 \rightarrow \text{have root } \alpha, \beta$$

$$\therefore a\alpha^2 + b\alpha + c = 0 \dots (i)$$

$$a\beta^2 + b\beta + c = 0 \dots (ii)$$

$$ax^2 - bx(x - 1) + c(x - 1)^2 = 0$$

divided by $(x - 1)^2$

$$a\left(\frac{x}{1-x}\right)^2 + b\left(\frac{x}{1-x}\right) + c = 0 \dots (iii)$$

From (i) and (iii) by transformation of roots

$$\alpha = \frac{x}{1-x} \Rightarrow \alpha - \alpha x = x \Rightarrow x = \frac{\alpha}{1+\alpha}$$

$$\text{so, roots are } \frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$$

11. **Ans (A)**

$$\text{Let } f(x) = -x^2 + ax + a$$



$$\therefore A. f(1) < 0$$

$$\Rightarrow (1) [-1 + a + a] < 0$$

$$\Rightarrow -1 + 2a < 0 \Rightarrow a < \frac{1}{2}$$

12. **Ans (C)**

$$\frac{x^2 - 3x - 10}{x^2 + x + 1} \leq 0 \quad \left\{ \begin{array}{l} \text{for } x^2 + x + 1 \\ a > 0 \text{ \& } D < 0 \end{array} \right\}$$

$$\therefore x^2 + x + 1 > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow x^2 - 3x - 10 \leq 0$$

$$\Rightarrow (x - 5)(x + 2) \leq 0$$



$$\text{Integral } x = -2, -1, 0, 1, 2, 3, 4, 5$$

$$\therefore \text{sum} = 12 \text{ Ans.}$$

13. Ans (A)

α & β are roots of $2x^2 + 3x - 1 = 0$

$$2\alpha^2 + 3\alpha - 1 = 0 \Rightarrow 2\alpha^{n+2} + 3\alpha^{n+1} - \alpha^n = 0$$

$$2\beta^2 + 3\beta - 1 = 0 \Rightarrow 2\beta^{n+2} + 3\beta^{n+1} - \beta^n = 0$$

By adding :

$$2(\alpha^{n+2} + \beta^{n+2}) + 3(\alpha^{n+1} + \beta^{n+1}) - (\alpha^n + \beta^n) = 0$$

$$\Rightarrow 2S_{n+2} + 3S_{n+1} - S_n = 0$$

$$\text{For } n = 5, 2S_7 + 3S_6 = S_5$$

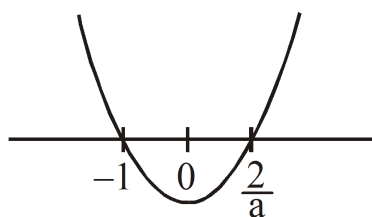
14. Ans (B)

$ax^2 + (a-2)x - 2 < 0$; for exactly two integral values of x .

$$x = \frac{-(a-2) \pm \sqrt{(a-2)^2 + 8a}}{2a} = \frac{-(a-2) \pm (a+2)}{2a}$$

$$= \frac{2}{a} \text{ or } -1$$

Case-1: $a > 0$... (1)



$\therefore$ Quadratic is -ve for exactly 2 integral values of x .

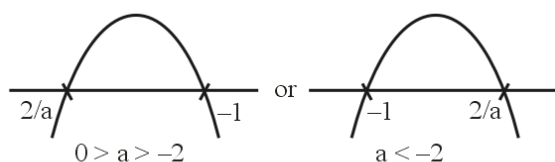
$$\frac{2}{a} \in (1, 2]$$

$$\frac{1}{a} \in \left(\frac{1}{2}, 1\right] \Rightarrow a \in [1, 2) \quad \dots (2)$$

$$(1) \cap (2)$$

$$a \in [1, 2) \quad \dots (3)$$

Case-2: $a < 0$... (4)



But in this case the Quadratic is not -ve for exactly 2 integral values of x

But in this case the Quadratic is -ve for more than 2 values of x

15. Ans (B)

We have

$$x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$$

$$\text{so } -3 < \frac{x^2 + mx + 1}{x^2 + x + 1} < 3$$

$$\Rightarrow -3(x^2 + x + 1) < x^2 + mx + 1 < 3(x^2 + x + 1)$$

$$\Rightarrow 4x^2 + (m+3)x + 4 > 0 \text{ and}$$

$$2x^2 + (3-m)x + 2 > 0 \text{ for real } x$$

$$D_1 < 0 \text{ and } D_2 < 0$$

$$\Rightarrow (m+3)^2 - 4 \times 4 \times 4 < 0 \text{ and}$$

$$(3-m)^2 - 4 \times 2 \times 2 < 0$$

$$\Rightarrow (m+3+8)(m+3-8) < 0 \text{ and}$$

$$(m-3+4)(m-3-4) < 0$$

$$\Rightarrow (m+11)(m-5) < 0 \text{ and}$$

$$(m+1)(m-7) < 0$$

$$\Rightarrow -11 < m < 5 \text{ and } -1 < m < 7$$

Taking intersection give $-1 < m < 5$

16. Ans (D)

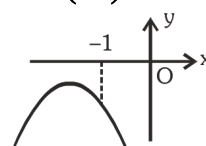
$$\sqrt{-x^2 + x + 6} \text{ is defined when } x^2 - x - 6 \leq 0.$$

$$\Rightarrow (x-3)(x+2) \leq 0.$$

$$\Rightarrow x \in [-2, 3] \text{ for these values of } x, x-4 < 0$$

and square root is ≥ 0 so all these x will be solution.

17. Ans (B)



$ax^2 + bx + c = 0$ has no real root

put $x = -1$

$$a - b + c < 0$$

from graph

$$a < 0 \text{ \& } c < 0$$

18. Ans (D)

$$\alpha + \beta \Rightarrow \frac{-b}{a} < 0 \Rightarrow \frac{b}{a} > 0$$

$\Rightarrow a$ and b are of same sign

$$\alpha\beta < 0 \Rightarrow \frac{c}{a} < 0$$

$\Rightarrow a$ and c are of opposite sign

19. Ans (A)

$$\frac{x^2 + 4x + 7}{x^2 + 4x + 5} = \frac{x^2 + 4x + 5}{x^2 + 4x + 5} + \frac{2}{x^2 + 4x + 5}$$

$$= 1 + \frac{2}{x^2 + 4x + 5} \quad \dots(1)$$

$$x^2 + 4x + 5 = (x+2)^2 + 1$$

$$\Rightarrow x^2 + 4x + 5 \in [1, \infty)$$

$$\Rightarrow \frac{2}{x^2 + 4x + 5} \in (0, 2]$$

$$\therefore \frac{x^2 + 4x + 7}{x^2 + 4x + 5} \in (1, 3]$$

20. Ans (A)

$$\text{Let } f(x) = x^2 - 5|x| + 1$$

$$\therefore x \in [-4, 2]$$

$$|x| \in [0, 4]$$

$$\text{Put } |x| = t \in [0, 4]$$

$$\therefore f(t) = t^2 - 5t + 1, t \in [0, 4]$$

$$f(0) = 1$$

$$f(4) = 16 - 5(4) + 1 = -3$$

$$f\left(\frac{5}{2}\right) = \frac{25}{4} - \frac{25}{2} + 1 = 1 - \frac{25}{4} = -\frac{21}{4}$$

$$\therefore f(t) \in \left[-\frac{21}{4}, 1\right]$$

$\therefore$ No. of integers is 7

MATHEMATICS

SEC-B

1. Ans (-12.00)

If equation true for all real values of x i.e., it is an identity

$$\left. \begin{aligned} a^3 - 3a - 2 &= 0 \Rightarrow a = -1, 2, -1 \\ a^2 - a - 2 &= 0 \Rightarrow a = -1 \& 2 \end{aligned} \right\} \cap \Rightarrow a = -1, 2$$

$$2a^4 - 10a + k = 0$$

$\Rightarrow$ Put $a = -1, 2$ in this

we get $k = -12$

2. Ans (1.25)

$$\text{Put } 2^x = t$$

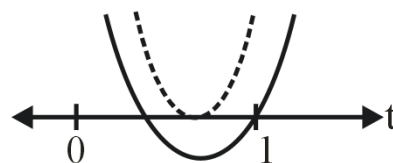
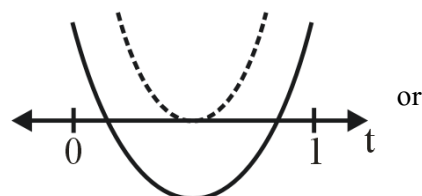
$$t^2 - (a-3)t + (a-4) = 0$$

$$\therefore x \leq 0$$

$$\Rightarrow 0 < 2^x \leq 1$$

$$\Rightarrow 0 < t \leq 1$$

Now apply roots of the eqn lie b/w (0, 1]



$$D \geq 0 \Rightarrow (a-3)^2 - 4(1)(a-4) \geq 0$$

$$\Rightarrow a^2 - 6a + 9 - 4a + 16 \geq 0$$

$$\Rightarrow a^2 - 10a + 25 \geq 0$$

$$\Rightarrow (a-5)^2 \geq 0$$

$$\Rightarrow a \in \mathbb{R} \quad \dots(1)$$

$$0 \leq \frac{-B}{2A} \leq 1 \Rightarrow 0 < \frac{a-3}{2} < 1$$

$$\Rightarrow 0 < a-3 \leq 2$$

$$\Rightarrow 3 < a \leq 5 \quad \dots(2)$$

$$A. f(0) > 0 \Rightarrow 1.(0+0+a-4) > 0$$

$$\Rightarrow a > 4 \quad \dots(3)$$

$$A. f(1) \geq 0 \Rightarrow 1(1-(a-3)+(a-4)) \geq 0$$

$$\Rightarrow 0 \geq 0$$

$$\therefore a \in \mathbb{R} \quad \dots(4)$$

$$(1) \cap (2) \cap (3) \cap (4)$$

$$a \in (4, 5] \therefore p=4, q=5$$

$$\Rightarrow \frac{q}{p} = \frac{5}{4} = 1.25$$

3. Ans (2.00)

$$x^2 + ax + a + 1 = 0$$

$$D = a^2 - 4(1)(a + 1)$$

$$\Rightarrow D = a^2 - 4a - 4$$

$$\Rightarrow D = a^2 - 4a + 4 - 8$$

$$\Rightarrow D = (a - 2)^2 - 8$$

For integral roots and $a \in \mathbb{I}$

Discriminant must be perfect square

$$\therefore 8 = (a - 2)^2 - D$$

Possible when

$$(a - 2)^2 = 9 \text{ and } D = 1$$

$$a - 2 = \pm 3$$

$$a = 5, -1$$

4. Ans (-1.00)

$$(x^2 + x - 2)(x^2 + x - 3) = 12$$

$$\text{Let } x^2 + x = t$$

$$(t - 2)(t - 3) = 12$$

$$t^2 - 5t - 6 = 0$$

$$t = -1, 6$$

$$\Rightarrow x^2 + x + 1 = 0 \text{ or } x^2 + x - 6 = 0$$

5. Ans (0.00)

$$y = \frac{2x + 3}{x + 4}, x > 1$$

$$x = \frac{3 - 4y}{y - 2}$$

$$\therefore x > 1 \Rightarrow \frac{3 - 4y}{y - 2} > 1$$

$$\Rightarrow \frac{3 - 5y + 2}{y - 2} > 0$$

$$\Rightarrow \frac{5(y - 1)}{(y - 2)} < 0 \Rightarrow y \in (1, 2)$$

6. Ans (0.00)

$$D > 0 \quad \forall x \in \mathbb{R}$$

$$8(2(a + b)^2 - (a^2 + 1)) > 0$$

$$\Rightarrow a^2 - (4b)a + 2b^2 - 1 > 0 \quad \forall a \in \mathbb{R}$$

$$D < 0$$

$$\Rightarrow 16b^2 - 4(2b^2 - 1) < 0$$

$$8b^2 + 4 < 0 \Rightarrow \text{no real solution}$$

7. Ans (8.00)

$$5 + 2\sqrt{6} = 5 + 2\sqrt{6} \times \frac{5 - 2\sqrt{6}}{5 - 2\sqrt{6}} = \frac{25 - 24}{5 - 2\sqrt{6}}$$

$$= \frac{1}{5 - 2\sqrt{6}}$$

$$\therefore \text{Let } (5 + 2\sqrt{6})^{x^2 - 3} = t$$

$$t + \frac{1}{t} = 10 \Rightarrow t^2 - 10t + 1 = 0$$

$$\Rightarrow t = \frac{10 \pm \sqrt{100 - 4}}{2} = 5 \pm 2\sqrt{6}$$

$$\therefore (5 + 2\sqrt{6})^{x^2 - 3} = 5 + 2\sqrt{6} \text{ or}$$

$$(5 + 2\sqrt{6})^{x^2 - 3} = 5 - 2\sqrt{6}$$

$$x^2 - 3 = 1 \quad x^2 - 3 = -1$$

$$x = \pm 2 \quad x = \pm \sqrt{2}$$

$$\therefore \text{Product} = (-2)(2)(-\sqrt{2})(\sqrt{2}) = 8$$

8. Ans (3.00)

$$a.f(4) < 0$$

$$\Rightarrow 1[16 - (3k - 1)4 + 5k] < 0$$

$$\Rightarrow -7k + 20 < 0$$

$$k > \frac{20}{7}$$

minimum integral value of k is 3.

9. Ans (6.00)

$$|x - 3| = t$$

$$t^2 - t - 2 = 0 \Rightarrow |x - 3| = -1 \text{ not possible}$$

$$\text{or } |x - 3| = 2 \Rightarrow x - 3 = \pm 2$$

$$x = 5, 1$$

10. Ans (0.00)

a, b, c are +ve real nos.

$$\underbrace{ax^2}_{+ve} + \underbrace{b|x|}_{+ve} + \underbrace{c}_{+ve} = 0$$

$$\therefore \text{LHS is always positive (i.e. } > 0)$$

$\therefore$ No real solution possible.

$\therefore$ Number of real roots = 0